Simulation Arena

# Discount Leakage & Cherry-Picking: The Basket Sabotage

**Cherry-pick risk 59/100 (Moderate): give this discount and there is a 63.2% chance the Support line or another offsetting line disappears, costing $156k - 31.9% of the deal's profit.**

Seed 42 | Run run_40b5007de13e | 2,000 scenarios tested | Cluster bootstrap resampling with empir… | 21 Sep 2026, 23:28

Same seed, same numbers - this run can be reproduced exactly.

## How this works

We went back through your past multi-line deals and found the ones where a line was discounted past 30%. In those deals we measured how often the other lines were cancelled inside 6 months, and compared it with how often those lines die anyway. That difference is the cherry-pick signal. We then rebuilt this quote 2,000 times, each time re-drawing whole past deals so the answer reflects how much history we actually have, and re-priced the basket with whatever survived. The risk score is three things you can check by hand: how far past your discount limit the ask sits (25%), how likely the offsetting lines are to vanish (50%), and how much of the profit sits in the line being cut (25%).

## The numbers that matter

| | | |
|---|---|---|
| CHERRY-PICK RISK SCORE<br>**59 / 100**<br>Built from three checkable pieces; the biggest one is how likely the offsetting lines are to vanish.<br>vs Moderate risk | PROFIT LIKELY TO LEAK AWAY<br>**$156k**<br>The profit this deal loses once you allow for how buyers actually behave.<br>vs 31.9% of the deal's profit | CHANCE THE SUPPORT LINE IS DROPPED<br>**63.2%**<br>How often the line that pays for the discount dies in the first 6 months.<br>vs 1.7x the normal 37.5% |
| CHANCE ANY OFFSETTING LINE DIES<br>**63.2%**<br>The odds the basket does not survive intact. | PROFIT EXPOSED ON THE SUPPORT LINE<br>**$95k**<br>$150k of profit with a 63.2% chance of walking. | PROFIT AT RISK<br>**31.9%**<br>Share of the bundle's profit that is exposed to cherry-picking. |
| PROFIT IF THEY KEEP EVERYTHING<br>**$489k**<br>The spreadsheet number: 54.3% margin on $900k. | PROFIT YOU SHOULD ACTUALLY PLAN FOR<br>**$333k**<br>The honest number after cherry-picking.<br>vs as low as $39k in a bad run | HOW MUCH MORE OFTEN THE WORST PAIR DIES<br>**1.7x**<br>Deals with a deep Software discount lose the Support line 1.7 times more often than usual.<br>vs 63.2% vs 37.5% |
| CHANCE THE BUNDLE ENDS UNDER YOUR 35% FLOOR<br>**10.2%**<br>How often the surviving basket is not worth signing at all. | | |

## How we got there

### 1. What is actually on the table

The basket is worth $900k and makes $489k if nothing is dropped. But 61.3% of that profit sits in Software - the exact line the buyer wants 35% off, which is 5 points past the discount line you set yourself.

Evidence: $300k of $489k profit in Software

**2. What history says happens next**

In 152 past deals where Software was discounted past 30%, the Support line was cancelled 63.2% of the time against a normal rate of 37.5% - 1.7 times more often. And they do not do it immediately - when a line dies it happens around month 3.

Evidence: source: your own history

**3. What the rebuild shows**

Rebuilding this quote 2,000 times, the profit lands at $333k on average, and in an unlucky run of twenty it is $39k or worse. There is a 10.2% chance the bundle that survives is worth less than your 35% margin floor.

Evidence: $156k average shortfall

**4. What to do about it**

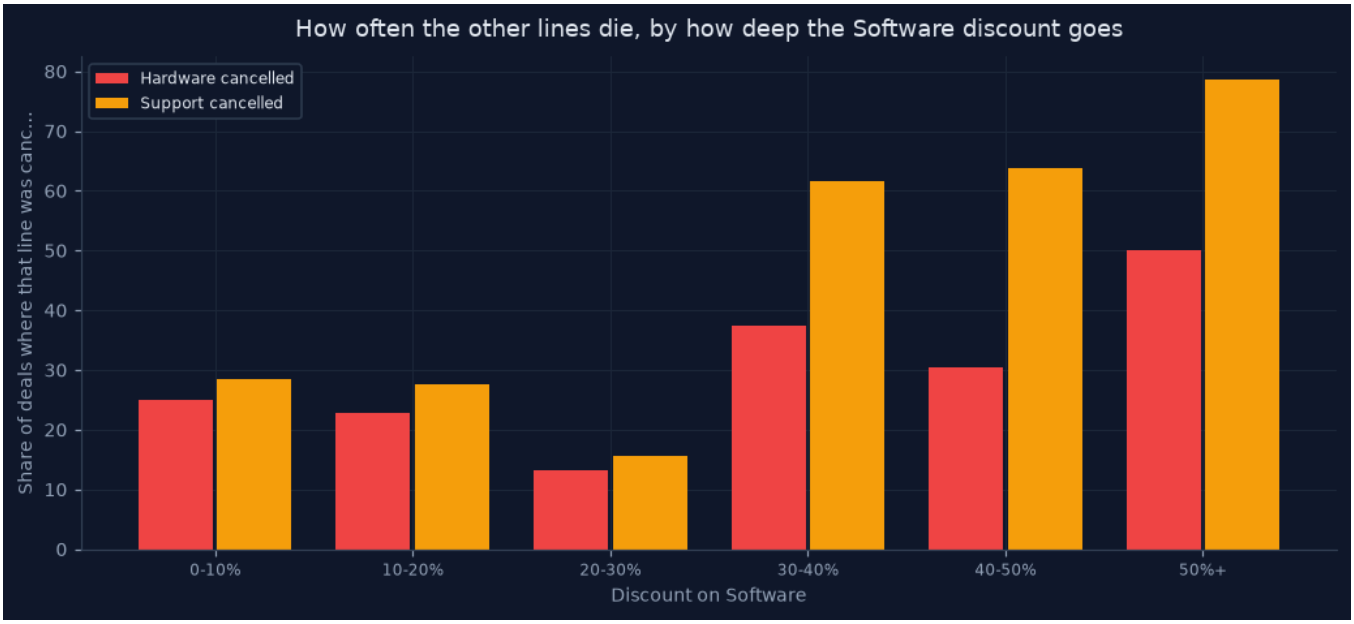
The biggest driver of the score is how likely the offsetting lines are to vanish. Tie the discount to the behaviour: if the Support line is not live at 80% of committed hours by month 6, the 35% Software discount comes off and Software reverts to standard pricing - worth up to $210k back.

Evidence: clawback up to $210k

## Charts

### How often the other lines die, by how deep the Software discount goes

Each bar is a group of past deals with a similar discount




*What this shows: If the bars jump once the discount passes your limit, buyers are not just buying more - they are buying differently. That jump is the cherry-pick signal.*

### Which discount poisons which line

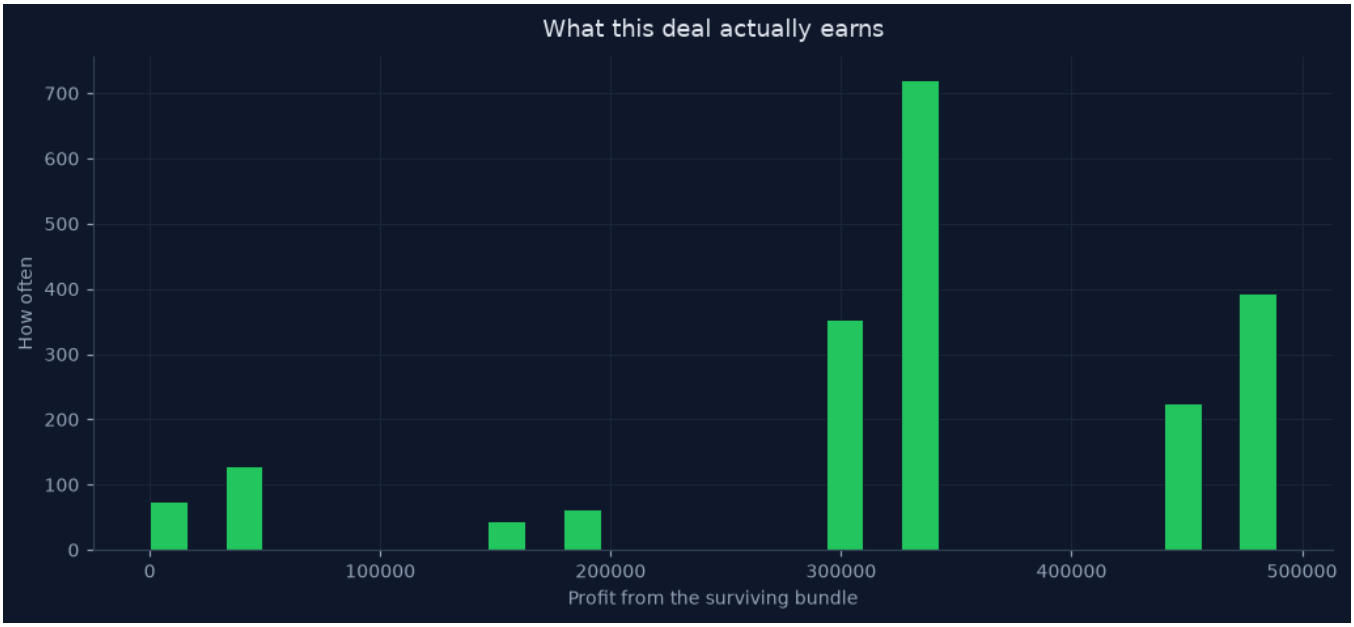
Numbers are 'times more likely than usual' when the row's line is discounted past 30% - rows with too little history are left out




*What this shows: Anything above 1.0 means the two events travel together. The hottest cell is the pair you should write a contract clause about.*

### What this deal actually earns

2,000 rebuilt bundles - the dashed line is the number on the quote




*What this shows: The gap between the tall bars and the 'if nothing is dropped' marker is the cost of cherry-picking. The left tail is the bad case you are signing up for.*

## Tables

### The quote, line by line

Where the profit sits, and how exposed each line is.

| Line | List price | Discount | Price after discount | Your margin | Profit | Share of profit | Chance it is dropped | Profit at risk |
|---|---|---|---|---|---|---|---|---|
| Software | $600,000 | 35.0% | $390,000 | 85.0% | $300,000 | 61.3% | 15.1% | $45,395 |
| Hardware | $300,000 | 5.0% | $285,000 | 18.0% | $39,000 | 8.0% | 36.2% | $14,112 |
| Support | $250,000 | 10.0% | $225,000 | 70.0% | $150,000 | 30.7% | 63.2% | $94,737 |

Row 1 carries the discount being asked for.

### What happens to each line when another line is discounted past 30%

Read across a row: 'when THIS line is deeply discounted, the chance THAT line dies'. A figure above 1.0 means the two events travel together.

| Discounted line (A) | Line that dies (B) | Deals like this | Normal rate | Rate after the discount | Times more often | Link strength |
|---|---|---|---|---|---|---|
| Hardware | Hardware | 20 | 25.2% | 40.0% | 1.58x | weak |
| Hardware | Services | 20 | 24.5% | 30.0% | 1.22x | none |
| Hardware | Software | 20 | 13.0% | 15.0% | 1.15x | none |
| Hardware | Support | 20 | 37.5% | 60.0% | 1.60x | weak |
| Services | Hardware | 24 | 25.2% | 16.7% | 0.66x | none |
| Services | Services | 24 | 24.5% | 16.7% | 0.68x | none |
| Services | Software | 24 | 13.0% | 8.3% | 0.64x | none |
| Services | Support | 24 | 37.5% | 33.3% | 0.89x | none |
| Software | Hardware | 152 | 25.2% | 36.2% | 1.43x | clear |
| Software | Services | 152 | 24.5% | 30.3% | 1.24x | weak |
| Software | Software | 152 | 13.0% | 15.1% | 1.16x | none |
| Software | Support | 152 | 37.5% | 63.2% | 1.68x | strong |

Rows where the second line dies far more often than normal are the cherry-pick signature.

### What the bundle actually earns

Every one of the 2,000 rebuilt bundles, summarised.

| Scenario | Profit | Margin | Versus the plan |
|---|---|---|---|
| If they keep everything | $489,000 | 54.3% | the number on the quote |
| Most likely outcome | $339,000 | 37.7% | -$150,000 |
| A bad run (1 in 20) | $39,000 | 4.3% | -$450,000 |
| Worst bundle we saw | $0 | 0.0% | -$489,000 |

Row 2 is the number to plan with.

### Does the buyer's tier change the behaviour?

Same test, split by customer tier (only tiers with enough history are shown).

| Customer tier | Deals | Deals with a deep discount | Chance each offsetting line dies |
|---|---|---|---|
| Enterprise | 107 | 46 | 37%, 59% |
| Mid-Market | 172 | 60 | 38%, 65% |
| SMB | 121 | 46 | 33%, 65% |

Use the worst tier as your planning number if you do not know the buyer well.

## What we recommend

**Approve the 35% only as a conditional discount - expect to lose $156k if you do not tie it to the Support line.**

**Risk Moderate** 59 out of 100 | Confidence: High

### Why

• 61.3% of this deal's profit sits in Software, the line being cut by 35%.

• Past deals like this one lose the offsetting lines 63.2% of the time (your own history).

• Expected profit after cherry-picking is $333k, not $489k - a $156k gap.

### What to do

**1.** Make the 35% Software discount conditional: if the Support line is not live at 80% of committed hours by month 6, Software pricing reverts to the standard tier (recovering up to $210k).

**2.** Do not let them drop one offsetting line and keep the other: make the discount conditional on the whole Hardware, Support basket surviving, not on any single line.

**3.** Put the test at month 6, not at signature - past buyers who walk do it around month 3.

**4.** There is a 10.2% chance the surviving bundle sits under your 35% floor - require CF sign-off above 30% on any single line.

**The trade-off:** Walking away from the 35% risks the deal; giving it unprotected costs about $156k. The clause lets you keep the deal and cap the loss. If the buyer refuses any condition, the mildest version is to cap Software at 30%, which still protects $30k.

## What we assumed and what to watch

### Worth knowing before you act

• Some category pairs rest on fewer than 30 past deals, so those figures are rough. The headline pair is the one with enough history behind it.

### Assumptions

• Quote under review: Software $600,000 at 35% off (85.0% margin), Hardware $300,000 at 5% off (18.0% margin), Support $250,000 at 10% off (70.0% margin).

• Deep discount means anything past 30%; a line counts as abandoned if it dies within 6 months.

• Abandonment rates come from your own history: 400 past deals, 203 deep-discount category observations.

• Risk score = 25% x how far past your discount limit the ask sits (10 points past = full marks) + 50% x chance an offsetting line dies + 25% x share of profit sitting in the discounted line.

• A cancelled line is treated as fully lost: no revenue and no cost.

• Bundle margin floor 35%; minimum support commitment 80% of hours.

• We replayed 2,000 bundles built from your own past deals; seed 42 - re-running gives identical numbers.

• Quote / deal ID -> your 'quote_id' column

• Customer tier / segment -> your 'customer_tier' column

• Line item ID -> your 'line_id' column

• Product line name -> your 'product_line' column

• Line category -> your 'category' column

• List price of the line -> your 'list_price' column

• Cost of the line -> your 'cost' column

• Discount on the line (%) -> your 'discount_pct' column

• Was the line cancelled? (1/0) -> your 'cancelled_within_6m' column

• Months until cancellation -> your 'months_to_cancel' column

## Appendix: the last steps of the run

Every line was produced while the model ran. Re-running with the same seed replays it exactly.

| Time | Stage | What happened |
|---|---|---|
| 0.00s | prepare | List price of the line -> your 'list_price' column |
| 0.00s | prepare | Cost of the line -> your 'cost' column |
| 0.00s | prepare | Discount on the line (%) -> your 'discount_pct' column |
| 0.00s | prepare | Was the line cancelled? (1/0) -> your 'cancelled_within_6m' column |
| 0.00s | prepare | Months until cancellation -> your 'months_to_cancel' column |
| 0.03s | prepare | Read 1,600 records into the model. |
| 0.03s | simulate | Running the simulation… |
| 0.03s | read | Reading the quote you want us to stress-test... |

| Time | Stage | What happened |
|---|---|---|
| 0.03s | read | The basket is worth $900k and carries $489k of profit (54.3% margin) if the buyer keeps every line. |
| 0.03s | read | The line being discounted hardest is Software at 35% off - that is 5 points past your 30% deep-discount line. |
| 0.03s | read | Learning from 400 past deals across 4 categories (Hardware, Services, Software, Support)... |
| 0.03s | prepare | Checking what happened in past deals that got a discount this deep... |
| 0.03s | prepare | Found it: when Software is discounted past 30%, the Support line is cancelled 63.2% of the time - 1.7x its normal rate. |
| 0.03s | prepare | Chance the offsetting lines disappear: 63.2% - that is what the discount is really buying. |
| 0.03s | simulate | Rebuilding 2,000 bundles that look like this one and re-pricing each of them... |
| 0.05s | simulate | Rebuilt 10% of the bundles... |
| 0.05s | simulate | Rebuilt 20% of the bundles... |
| 0.05s | simulate | Rebuilt 30% of the bundles... |
| 0.05s | simulate | Rebuilt 40% of the bundles... |
| 0.05s | simulate | Rebuilt 60% of the bundles... |
| 0.05s | simulate | Rebuilt 70% of the bundles... |
| 0.05s | simulate | Rebuilt 80% of the bundles... |
| 0.05s | simulate | Rebuilt 90% of the bundles... |
| 0.11s | summarize | All bundles priced. Working out what this deal really earns... |
| 0.11s | summarize | Done. Cherry-pick risk 59/100 (Moderate) - expect to lose $156k unless the discount is made conditional. |